

keytheorems package

version 0.3.0dev

github.com/mbertucci47/keytheorems

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Abstract

An expl3-implementation of a key-value interface to `amsthm`, implementing most of the functionality provided by `thmtools`. Several issues encountered with `thmtools` are avoided (see the README for a list) and a few new features are added.

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1 Dependencies

The package depends on the `aliascnt`, `amsthm`, `refcount`, and `translations` packages. The `tcolorbox`^{→P.13} and `tcolorbox-no-titlebar`^{→P.13} keys require `tcolorbox`, and the `numbered=unless-unique`^{→P.8} key requires the `unique` package. A $\text{\LaTeX}$ kernel no older than 2023-06-01 is required; if older than 2024-06-01, `nameref` is required.

2 Global options

`\keytheoremset{<options>}`

Every key in this section can be given as an option to `\usepackage` or in `\keytheoremset`, with the exception that `continues-code`^{→P.3} can only be used in the latter.

2.1 Compatibility options

`overload` (initially unset)

Redefines `\newtheorem` to internally use the `keytheorems` machinery. The syntax remains the same. This is automatically set by `thmtools-compat`.

`thmtools-compat` (initially unset)

For compatibility with `thmtools` syntax. For most documents,

```
\usepackage[thmtools-compat]{keytheorems}
```

should be a drop-in replacement for `\usepackage{amsthm,thmtools}`. The option defines the commands in the left column below. The right column lists the corresponding `keytheorems` replacement that should be used in new documents.

thmtools command		keytheorems replacement
<code>\declaretheorem</code>	→	<code>\newkeytheorem</code> ^{→P.4}
<code>\declaretheoremstyle</code>	→	<code>\newkeytheoremstyle</code> ^{→P.14}
<code>\listoftheorems</code>	→	<code>\listofkeytheorems</code> ^{→P.17}
<code>\listtheoremname</code>	→	<code>title</code> ^{→P.19} key
<code>\addtotheoremheadhook</code>		
<code>\addtotheoremstylehook</code>	→	<code>\addtotheoremhook</code> ^{→P.21}
<code>\addtotheoremstylefoothook</code>		
<code>\addtotheoremstylepostfoothook</code>		
<code>restatable</code> environment	→	<code>store</code> ^{→P.6} key
<code>restatable*</code> environment	→	<code>store*</code> ^{→P.6} key

Also defined are the `shaded` and `thmbox` keys, implemented internally with `tcolorbox` rather than the `shadethm` and `thmbox` packages, respectively.

2.2 Other global options

`auto-translate=true|false` (default `true`, initially `true`)

If `false`, `keytheorems` does not automatically translate the title text used for `\listofkeytheorems`^{→P.17} and the note produced by the `continues`^{→P.5} key. These texts can be manually customized with the `title`^{→P.19} and `continues-code`^{→P.3} keys, respectively.

`continues-code`= $\langle$ *code with #1* $\rangle$ (initially `\GetTranslation{keythms_continues}\pageref{#1}`)

The code used to typeset the note produced by the `continues`^{→P.5} key. If English or an unknown language is used, defaults to `continuing from p.\, \pageref{#1}`. Currently (likely inaccurate!) translations exist for several European languages.

`predefined`= $\{\langle$ *options* $\rangle\}$ (initially unset)

This is a convenience key, similar to `ntheorem`'s `standard` option, that predefines a set of theorems that, unless `auto-translate`^{→P.2} is set to `false`, are translated into the current language if translations exist. The predefined theorems are

- plain style: `conjecture`, `corollary`, `lemma`, `proposition`, `theorem`;
- definition style: `axiom`, `definition`, `example`;
- remark style: `remark`.

If your language does not have translations, please feel free to open a GitHub pull request.

These theorems are provided at the end of the preamble (specifically, in the `begindocument` hook) with `\providekeytheorem`^{→P.4} so will not overwrite user-defined environments with the same name. By default, the predefined theorems share a counter and do not have a parent counter. These settings can be changed by calling `siblings=false` and `parent=\langle counter \rangle`, respectively, in $\langle$ *options* $\rangle$.

```
\usepackage[
  predefined={parent=section}
]{keytheorems}

% or equivalently
\usepackage{keytheorems}
\keytheoremset{predefined={parent=section}}
```

`qed-symbol`= $\langle$ *symbol* $\rangle$ (initially `\openbox`)

Redefines `\qedsymbol` to be $\langle$ *symbol* $\rangle$.

`restate-counters`= $\{\langle$ *comma-list of counters* $\rangle\}$ (initially `{equation}`)

Additional counters whose values are preserved when a theorem is restated. This key does not reset the list, so you don't need to include `equation` in $\langle$ *comma-list* $\rangle$.

`store-all` (initially unset)

Tells `keytheorems` to grab the body of each theorem so it can later be printed with the `print-body`^{→P.20} option of `\listofkeytheorems`^{→P.17}. Note that this means a theorem body *cannot* contain verbatim material.

`store-sets-label` (initially unset)

Defines the `store`^{→P.6} key to also set `label`^{→P.5}, i.e. it makes `store=\langle tag \rangle` equivalent to `store=\langle tag \rangle, label=\langle tag \rangle`. Similarly for `store*`^{→P.6}.

3 Defining theorems

`\newkeytheorem{⟨env name⟩}[⟨options⟩]`

Defines a theorem environment $\langle env\ name \rangle$ which itself takes a few options (see subsection 3.1). You can also declare multiple theorems at once by replacing $\langle env\ name \rangle$ with a comma-list of names, e.g.

`\newkeytheorem{theorem,lemma,proposition}[⟨options⟩].`

By default, the theorem's printed name is a title-cased $\langle env\ name \rangle$. This can be changed with the `name`^{→P.8} key. All $\langle options \rangle$ are described in subsections 3.2 and 3.3.

```
% preamble
\newkeytheorem{theorem}

% document
\begin{theorem}
There are infinitely many prime numbers.
\end{theorem}
```

Theorem 1. *There are infinitely many prime numbers.*

`\renewkeytheorem{⟨env name⟩}[⟨options⟩]`

`\providekeytheorem{⟨env name⟩}[⟨options⟩]`

`\declarekeytheorem{⟨env name⟩}[⟨options⟩]`

Sometimes a package or class defines theorems that need to be overwritten by the user. For this case, `keytheorems` provides `\renewkeytheorem` which redefines $\langle env\ name \rangle$ or errors if it is not defined. For completeness, also provided are `\providekeytheorem` and `\declarekeytheorem`. The former only defines $\langle env\ name \rangle$ if it is not already defined; the latter always overwrites $\langle env\ name \rangle$.

3.1 Keys available to theorem environments

As in `amsthm`, theorems can take an optional argument that contains a note or heading.

```
\begin{theorem}[Bertrand's postulate]
For every  $n \geq 1$ , there is a prime number  $p$  with  $n < p \leq 2n$ .
\end{theorem}
```

Theorem 2 (Bertrand's postulate). *For every $n \geq 1$, there is a prime number p with $n < p \leq 2n$.*

Alternatively, the optional argument may contain any of the following keys.

`note=⟨text⟩` (initially unset)

Alias `name`. This is the key-value equivalent of the optional argument described above. This syntax, however, allows the argument to contain other keys.

```
\begin{theorem}[note=Legendre's formula]
The number  $n!$  contains the prime factor  $p$  exactly
 $\lfloor \sum_{k \geq 1} \lfloor \frac{n}{p^k} \rfloor \rfloor$ 
\end{theorem}
```

```
times.
\end{theorem}
```

Theorem 3 (Legendre’s formula). *The number $n!$ contains the prime factor p exactly*

$$\sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor$$

times.

`short-note`=*(text)* (initially unset)

Alias `short-name`. This replaces the value of `note`^{→P.4} when displayed in the list of theorems (`\listofkeytheorems`^{→P.17}).

`label`=*(label name)* (initially unset)

This is the key-value equivalent of `\begin{theorem} \label{(label name)}`.

```
\begin{theorem}[label=bezout,note=Bézout's identity]
Let  $a$  and  $b$  be integers. Then there exist integers  $x$  and  $y$ 
such that  $ax+by=\gcd(a,b)$ .
\end{theorem}
See \zcref{bezout}.
```

Theorem 4 (Bézout’s identity). *Let a and b be integers. Then there exist integers x and y such that $ax + by = \gcd(a, b)$.*

See theorem 4.

`manual-num`=*(text)* (initially unset)

Use this to override the printed number of a theorem. It is useful for making “starred” versions of other theorems, perhaps to represent a reformulated or more difficult version.

```
\begin{theorem}[manual-num=\ref*{bezout}*]
Let  $a_1, \dots, a_n$  be integers. Then there exist integers
 $x_1, \dots, x_n$  such that  $a_1x_1 + \dots + a_nx_n = \gcd(a_1, \dots, a_n)$ .
\end{theorem}
\begin{theorem}[manual-num=\faRocket] % requires fontawesome5
Don't confuse your readers by changing the numbering without good
reason.
\end{theorem}
```

Theorem 4*. *Let $a_1, \dots, a_n$ be integers. Then there exist integers $x_1, \dots, x_n$ such that $a_1x_1 + \dots + a_nx_n = \gcd(a_1, \dots, a_n)$.*

Theorem 🚀. *Don’t confuse your readers by changing the numbering without good reason.*

`continues*`=*(label name)* (initially unset)

Pick up a theorem where you left off. The theorem number remains the same. The printed text can be customized with the `continues-code`^{→P.3} option. The starred version also copies the theorem `note`^{→P.4} and `short-note` if they are nonempty.

```

\begin{theorem}[continues=bezout]
Moreover, the integers of the form  $az+bt$  are exactly the multiples
of  $\gcd(a,b)$ .
\end{theorem}
\begin{theorem}[continues*=bezout]
Moreover, the integers of the form  $az+bt$  are exactly the multiples
of  $\gcd(a,b)$ .
\end{theorem}

```

Theorem 4 (continuing from p. 5). *Moreover, the integers of the form $az+bt$ are exactly the multiples of $\gcd(a,b)$.*

Theorem 4 (Bézout's identity, continuing from p. 5). *Moreover, the integers of the form $az + bt$ are exactly the multiples of $\gcd(a,b)$.*

`store*=<tag>` (initially unset)

Alias `restate*`. Stores the the theorem to be restated at any point in the document with `\getkeytheorem`^{P.16}. With the starred version, counters and labels are taken from the copy called with `\getkeytheorem`, so in this case can only be restated once. This allows you, for example, to write all theorems and proofs in the appendix and call `\getkeytheorem` at the appropriate time mid-document. For the numbering to be correct, the unstarred key will need at most two runs and the starred key at most three runs.

```

\begin{theorem}[store=blub]
A theorem worth restating.
\end{theorem}
More brilliant mathematics.
\getkeytheorem{blub}

```

Theorem 5. *A theorem worth restating.*

More brilliant mathematics.

Theorem 5. *A theorem worth restating.*

A theorem given this key *cannot* contain verbatim material or other unexpected catcodes such as a `tikz-cd` diagram. The latter issue can be averted with the `ampersand-replacement` key.

```

% preamble
\usepackage{tikz}
\usetikzlibrary{cd}

% document
\begin{lemma}[store=fiberprod]
For any  $S$ -schemes  $X$  and  $Y$ , there exists a scheme  $X \times_S Y$ 
with morphisms to  $X$  and  $Y$  such that the diagram
\begin{tikzcd}[ampersand replacement=\&]
X \times_S Y \ar[r] \ar[d] \& X \ar[d] \\
Y \ar[r] \& S
\end{tikzcd}
commutes and is universal with respect to this property.

```

```

\end{lemma}
\dots
\getkeytheorem{fiberprod}

```

Lemma 6. *For any S -schemes X and Y , there exists a scheme $X \times_S Y$ with morphisms to X and Y such that the diagram*

$$\begin{array}{ccc}
 X \times_S Y & \longrightarrow & X \\
 \downarrow & & \downarrow \\
 Y & \longrightarrow & S
 \end{array}$$

commutes and is universal with respect to this property.

...

Lemma 6. *For any S -schemes X and Y , there exists a scheme $X \times_S Y$ with morphisms to X and Y such that the diagram*

$$\begin{array}{ccc}
 X \times_S Y & \longrightarrow & X \\
 \downarrow & & \downarrow \\
 Y & \longrightarrow & S
 \end{array}$$

commutes and is universal with respect to this property.

restate-keys= $\{\langle \text{list of keys} \rangle\}$ (initially unset)

Allows passing different keys to the restated theorem. At the moment this is only useful with the `noteP.4` key.

```

\begin{theorem}[
  store=rktest,
  note=Original,
  restate-keys={note=Restated}
]
Wow, yet another theorem.
\end{theorem}
\getkeytheorem{rktest}

```

Theorem 7 (Original). *Wow, yet another theorem.*

Theorem 7 (Restated). *Wow, yet another theorem.*

listhack=`true|false` (initially `false`)

Meant only to be used with the `breakP.15` style key for a theorem starting with a list. Compare:

```

% preamble
\newkeytheoremstyle{breaksty}{break}
\newkeytheorem{observation}[style=breaksty]

% document
\begin{observation}

```

```

\begin{enumerate}
\item First item
\item Second item
\end{enumerate}
\end{observation}

\begin{observation}[listhack=true]
\begin{enumerate}
\item First item
\item Second item
\end{enumerate}
\end{observation}

```

Observation 1. 1. *First item*

 2. *Second item*

Observation 2.

 1. *First item*

 2. *Second item*

Note that the value `true` must be explicitly set so that `listhack` is not interpreted as the note text.

`seq=<name>` (initially unset)

Adds the theorem to a custom sequence `<name>` that can then be listed with `\listofkeytheorems[seq=<name>]`. See `seq` ^{P.20} for more details.

3.2 Keys also defined in thmtools

These are the [`<options>`] available to `\newkeytheorem`. Except for `name` and `style` ^{P.9}, each key below can also be used in `\newkeytheoremstyle` ^{P.14}. For more description, see the `thmtools` package.

`name=<display name>` (initially title-cased `<env name>`)

Aliases `heading` and `title`.

```

% preamble
\newkeytheorem{mythm}[name=Some Name]

% document
\begin{mythm}
Some text
\end{mythm}

```

Some Name 1. *Some text*

`numbered=true|false|unless-unique` (default `true`, initially `true`)

For compatibility with `thmtools`, also accepts the values `yes`, `no`, and `unless unique`.

```
% preamble
\newkeytheorem{theorem*}[name=Theorem, numbered=false]

% document
\begin{theorem*}
An unnumbered theorem.
\end{theorem*}
```

Theorem. *An unnumbered theorem.*

parent= $\langle counter \rangle$ (initially unset)

Aliases **numberwithin** and **within**.

```
% preamble
\newkeytheorem{conjecture}[parent=section]

% document
\begin{conjecture}
The first number is the section.
\end{conjecture}
```

Conjecture 3.1. *The first number is the section.*

sibling= $\langle counter \rangle$ (initially unset)

Aliases **numberlike** and **sharenumber**.

```
% preamble
\newkeytheorem{lemma}[sibling=theorem]

% document
\begin{lemma}
This shares its counter with \texttt{theorem}.
\end{lemma}
```

Lemma 8. *This shares its counter with **theorem**.*

style= $\langle style name \rangle$ (initially unset)

Accepts any $\langle style name \rangle$ defined by `\newkeytheoremstyle`^{P.14}, as well as any of the predefined amsthm styles: **plain**, **definition**, and **remark**.

```
% preamble
\newkeytheorem{remark}[style=remark]

% document
\begin{remark}
It's nice to distinguish remarks from definitions and theorems.
\end{remark}
```

Remark 1. It's nice to distinguish remarks from definitions and theorems.

preheadhook= $\langle code \rangle$ (initially unset)

postheadhook= $\langle code \rangle$ (initially unset)

prefoothook= $\langle code \rangle$ (initially unset)

`postfoothook=<code>` (initially unset)

Details in [section 7](#).

```
% preamble
\newkeytheorem{test}[
  preheadhook=PREHEAD,
  postheadhook=POSTHEAD,
  prefoothook=PREFOOT,
  postfoothook=POSTFOOT
]
```

```
% document
\begin{test}
Some text
\end{test}
```

PREHEAD

Test 1. *POSTHEAD*Some text *PREFOOT*

POSTFOOT

`qed=<symbol>` (default `\qedsymbol`, initially unset)

Adds `<symbol>` to the end of the theorem body. If no value is given, current value of `\qedsymbol` is used (one can redefine this or set it with `qed-symbolP.3`). By default, this is $\square$.

```
% preamble
\newkeytheorem{example}[qed]
\newkeytheorem{solution}[qed=$\clubsuit$]

% document
\begin{example}
Some text.
\end{example}
\begin{solution}
Some more text.
\end{solution}
```

Example 1. *Some text.*

$\square$

Solution 1. *Some more text.*

$\clubsuit$

`refname=<ref name> or {<singular name>,<plural name>}` (initially `\MakeLowercase <display name>`)

If a single string, then the name used by `hyperref`'s `\autoref`, `cleveref`'s `\cref`, and `zref-clever`'s `\zceref`. If two strings separated by a comma, then the second string is the plural form used by `\cref`.

`Refname=<ref name> or {<singular name>,<plural name>}` (initially `\MakeUppercase <display name>`)

Same as `refname` but for `\Autoref`, `\Cref`, and `\zceref` with any of its capitalizing options. Note that `\Autoref` is defined by `keytheorems`, but requires `hyperref` to work. As with `\autoref`, there is also a starred version `\Autoref*` that suppresses the hyperlink.

```

% preamble
\newkeytheorem{prop}[
  name=Proposition,
  refname={proposition,propositions},
  Refname={Proposition,Propositions}
]

% document
\begin{prop}[label=abc]
Some text.
\end{prop}
\begin{prop}[label=def]
Some more text.
\end{prop}
Consider \zcref{abc,def}. \Autoref{abc} \dots

```

Proposition 1. *Some text.*

Proposition 2. *Some more text.*

Consider propositions 1 and 2. Proposition 1 ...

Both `cleveref` and `zref-clever` define default reference names for some commonly used counters like `theorem`, `lemma`, etc. For technical reasons, unless explicit values for `refname`^{→P.10} and `Refname`^{→P.10} are given, `keytheorems` does not try to change these defaults at all in the case of `cleveref` and only the singular name in the case of `zref-clever`. The easiest way to get exactly the output you want is to just explicitly set `refname`^{→P.10} and `Refname`^{→P.10}.

! The `cleveref`^{→CTAN} package has not been updated since 2018 and contains several incompatibilities with the $\text{\LaTeX}$ kernel. These are often patched by the $\text{\LaTeX}$ team, but further incompatibilities are likely to arise with each future update. For this reason, I recommend moving to `zref-clever`^{→CTAN}. It offers all the same features as `cleveref` and is actively maintained.

3.3 Keys added by `keytheorems`

`counter-format`=*<code>* (initially unset)

Syntactic sugar that essentially does `\renewcommand{\the<counter>}{<code>}`. The *<code>* should not contain any unexpandable tokens such as formatting commands. Formatting should be taken care of in the style keys `headfont`^{→P.15} and `numberfont`^{→P.16}. If used with an unnumbered theorem, a warning is issued.

```

% preamble
\newkeytheorem{mainthm}[
  name=Theorem,
  counter-format=\Alph{mainthm},
]

% document
\begin{mainthm}
The first main result, distinguished by using letters.
\end{mainthm}
\begin{mainthm}

```

And here is the second main result.
`\end{mainthm}`

Theorem A. *The first main result, distinguished by using letters.*

Theorem B. *And here is the second main result.*

Eventually L^AT_EX will allow syntax of the form `\Alph*` similar to `enumitem`'s `label` key, where the `*` means “use the current counter” (see [latex2e#1632](#)). Then the above example could be written as `counter-format=\Alph*`.

`leftmargin=<length>`
`rightmargin=<length>`
`margin=<length>` (initially 0pt)

Sets the left (respectively, right) margin of the theorem relative to the text width. The `margin` key sets both simultaneously. This sets the theorem apart from the text, similar to a block quote. The code was adapted from Enrico Gregorio's T_EX Stack Exchange answers:

- [How to change margins in enunciation \(theorem-like environment\)?](#)
- [A theoremstyle with complete indentation using amsthm](#)

```
% preamble
\newcommand{\marginthmtext}{%
  We need some text to show off theorems with margins. }
\newkeytheorem{quotethm}[name=Quote Theorem, margin=1cm]
\newkeytheorem{indentedthm}[name=Indented Theorem, leftmargin=1cm]

% document
\marginthmtext\marginthmtext\marginthmtext

\begin{quotethm}
\marginthmtext\marginthmtext\marginthmtext
\end{quotethm}

\marginthmtext\marginthmtext\marginthmtext

\begin{indentedthm}
\marginthmtext\marginthmtext\marginthmtext
\end{indentedthm}
```

We need some text to show off theorems with margins. We need some text to show off theorems with margins. We need some text to show off theorems with margins.

Quote Theorem 1. *We need some text to show off theorems with margins. We need some text to show off theorems with margins. We need some text to show off theorems with margins.*

We need some text to show off theorems with margins. We need some text to show off theorems with margins. We need some text to show off theorems with margins.

Indented Theorem 1. *We need some text to show off theorems with margins. We need some text to show off theorems with margins. We need some text to show off theorems with margins.*

`tcolorbox={\tcolorbox options}` (initially unset)

This key specifies that the theorem be placed inside a `tcolorbox` environment with `\tcolorbox options`. The theorem head is typeset as a `tcolorbox` title; to avoid this see `tcolorbox-no-titlebar`.

```
% preamble
\tcbset{
  defstyle/.style={
    arc=0mm,
    colback=blue!5!white,
    colframe=blue!75!black
  },
}
\newkeytheorem{corollary}[tcolorbox]
\newkeytheorem{definition}[style=definition, tcolorbox={defstyle}]

% document
\begin{corollary}
Products exist in the category of schemes over  $S$ .
\end{corollary}
\begin{definition}[Dedekind domains]
A \emph{Dedekind domain} is an integrally closed, Noetherian domain of
dimension one.
\end{definition}
```

Corollary 1.

Products exist in the category of schemes over S .

Definition 1 (Dedekind domains).

A *Dedekind domain* is an integrally closed, Noetherian domain of dimension one.

`tcolorbox-no-titlebar={\tcolorbox options}` (initially unset)

Same usage as `tcolorbox` but the theorem head is typeset as usual, not as a `tcolorbox` title.

```
% preamble
\newkeytheorem{boxcor}[
  tcolorbox-no-titlebar={colback=red!10},
  name=Corollary, sibling=corollary
]

% document
\begin{boxcor}[Cauchy's theorem]
Let  $G$  be a finite group and  $p$  a prime dividing the order of  $G$ .
Then  $G$  contains an element of order  $p$ .
\end{boxcor}
```

Corollary 2 (Cauchy’s theorem). *Let G be a finite group and p a prime dividing the order of G . Then G contains an element of order p .*

tcolorbox offers its own comprehensive `theorems` library. If all of your theorems are to be tcolorboxes, I highly recommend using it instead of this package! However, if only some of your theorems will use a tcolorbox, you may want to replicate the styles of `\NewTcbTheorem`. Here is an example that emulates tcolorbox’s `standard` theorem style.

```
% preamble
\tcbset{
  thmstyle/.style={
    colback=green!5,
    colframe=green!35!black},
}
\newkeytheoremstyle{tcb-standard}{
  tcolorbox=thmstyle,
  headpunct={},
  notebraces={}{},
  noteseparator={: },
  notefont=\bfseries,
  bodyfont=\normalfont,
}
\newkeytheorem{mytheo}[
  name=Theorem,
  style=tcb-standard
]

% document
\begin{mytheo}[Quillen-Suslin]
Every finitely generated projective module over a polynomial ring is free.
\end{mytheo}
```

Theorem 1: Quillen-Suslin

Every finitely generated projective module over a polynomial ring is free.

4 Theorem styles

`\newkeytheoremstyle`{ $\langle name \rangle$ }{ $\langle options \rangle$ }

This is keytheorems’ version of thmtools’ `\declaretheoremstyle`. Since it makes little sense to define a style with no keys, we’ve made the $\langle options \rangle$ argument mandatory. The defined style can be used with either the `style`^{P.9} key or the traditional `\theoremstyle`. Note that unlike amsthm’s `\newtheoremstyle`, this command will error if a style has already been defined.

`\renewkeytheoremstyle`{ $\langle env name \rangle$ }{ $\langle options \rangle$ }

`\providekeytheoremstyle`{ $\langle env name \rangle$ }{ $\langle options \rangle$ }

`\declarekeytheoremstyle`{*<env name>*}{*<options>*}

To overwrite an existing style, there is the analogous `\renewkeytheoremstyle`. For completeness, also provided are `\providekeytheoremstyle` and `\declarekeytheoremstyle`.

4.1 Keys also defined in thmtools

The following keys have the same meaning and syntax as the corresponding thmtools keys. In addition to the list below, most of the keys available to `\newkeytheorem`^{P.4} can be used in `\newkeytheoremstyle`.

`bodyfont`=** (initially `\itshape`)

`break` (initially unset)

Do not use this with the `postheadspace` key.

`headfont`=** (initially `\bfseries`)

`headformat`=`margin`|`swapnumber`|*<code using \NAME, \NUMBER, and \NOTE>*

Alias `headstyle`. Within *<code>*, the commands `\NAME`, `\NUMBER`, and `\NOTE` correspond to the formatted parts of the theorem head.

! In `headformat`, you may also use the traditional amsthm commands `\thmname`, `\thmnumber`, and `\thmnote`, where #1 is the theorem name, #2 the number, and #3 the note. `keytheorems` expands the head spec inside `\text_expand:n` so for these commands to work properly, the package adds them to `\l_text_expand_exclude_tl`. Note also that if you use these lower-level commands, the style keys `notebraces`, `notefont`, `noteseparator`^{P.16}, and `numberfont`^{P.16} will have no effect (of course, you can manually control these things inside the commands' arguments).

`headindent`=*<length>* (initially `0pt`)

`headpunct`=*<code>* (initially `{.}`)

`notebraces`=*<{<left brace>}<right brace>* (initially `{(}{)}`)

`notefont`=** (initially `\fontseries\mddefault\upshape`)

`postheadspace`=*<skip expr>* (initially `5pt plus 1pt minus 1pt`)

Do not use this with the `break` key.

`spaceabove`=*<skip expr>* (initially `\topsep`)

`spacebelow`=*<skip expr>* (initially `\topsep`)

! With `tcolorbox`^{P.13} and `tcolorbox-no-titlebar`^{P.13}, the `spaceabove` and `spacebelow` keys are internally passed to `tcolorbox`'s `before skip` and `after skip`. When no explicit `spaceabove` or `spacebelow` values are given, `tcolorbox` defaults are used instead of `\topsep`.

4.2 Keys added by keytheorems

`inherit-style`= $\langle style\ name \rangle$ (initially unset)

Inherit the keys of any style declared with `\newkeytheoremstyle`^{P.14}. Additionally, the three styles predefined by amsthm are possible values: `plain`, `definition`, and `remark`.

`noteseparator`= $\langle code \rangle$ (initially $\sqcup$)

The code inserted before the note, and printed only if there is a note. This is executed *before* the font commands set by `notefont`^{P.15} take effect.

`numberfont`= $\langle font\ declarations \rangle$ (initially `\upshape`)

For almost all theorem styles, it is recommended that you *do not* change this setting.

!

For the AMS classes `amsart`, `amsbook`, and `amsproc`, as well as the `amsart`-based `acmart` and `aomart`, the initial key values are slightly different those listed in sections 4.1 and 4.2 in order to match those class's defaults. See subsection 8.2 for details.

5 Restating theorems

When a theorem is given the `store`^{P.6} key, the contents of the theorem are saved and written to a `.thlist` file. At the start of the next run, this file is input at the beginning of the document and allows you to retrieve the stored theorems at any point, before or after the original theorem.

`\getkeytheorem`[$\langle property \rangle$]{ $\langle tag \rangle$ }

Retrieves the theorem given the key `store`= $\langle tag \rangle$ or `store*`= $\langle tag \rangle$. An optional $\langle property \rangle$ can be given to retrieve only the corresponding part of the theorem. Currently only the property `body` is implemented, which retrieves the (unformatted) body of the theorem.

```
\getkeytheorem{mytag}

\begin{example}[store=mytag]
Fascinating example.
\end{example}

\getkeytheorem[body]{mytag}
```

Example 2. *Fascinating example.* $\square$

Example 2. *Fascinating example.* $\square$

Fascinating example.

`\IfRestatingTF`{ $\langle true\ code \rangle$ }{ $\langle false\ code \rangle$ }

`\IfRestatingT`{ $\langle true\ code \rangle$ }

`\IfRestatingF`{ $\langle false\ code \rangle$ }

Executes $\langle true\ code \rangle$ if being retrieved with `\getkeytheorem` and $\langle false\ code \rangle$ if in the original theorem. This is reversed if `store*` is used.

```

\begin{example}[store=hmm]
I am the \IfRestatingTF{restated}{original} example!
\end{example}

\getkeytheorem{hmm}

```

Example 3. *I am the original example!* □

Example 3. *I am the restated example!* □

5.1 Restating theorems from an external file

`\externaltheorems[⟨prefix⟩]{⟨file name⟩}`

This is `keytheorems`' version of the `xr` package's `\externaldocument`. It allows the user to restate theorems from another document's `.thlist` file. Say you have a file `mycoolpaper.tex`,

```

% mycoolpaper.tex
\documentclass{article}
\usepackage{keytheorems}
\newkeytheorem{theorem}
\begin{document}
\begin{theorem}[store=cooltheorem]
My cool theorem.
\end{theorem}
\end{document}

```

and you'd like to restate the theorem with tag `cooltheorem` in another file `myothercoolpaper.tex` with the same numbering as in the original paper. Since your new paper probably also has cool theorems that you may want to tag as `cooltheorem`, you'd like to give all restatable theorems from `mycoolpaper.tex` a prefix when retrieved with `\getkeytheorem`^{P.16}, say "`orig:`". Just call `\externaltheorems[orig:]{mycoolpaper}` after loading `keytheorems` in the new document. Then any stored theorem from `mycoolpaper.tex` can be retrieved with `\getkeytheorem{orig:⟨tag⟩}`.

```

% myothercoolpaper.tex
\documentclass{article}
\usepackage{keytheorems}
\externaltheorems[orig:]{mycoolpaper}
\newkeytheorem{theorem}
\begin{document}
\getkeytheorem{orig:cooltheorem}
\end{document}

```

It is important that the `theorem` environment is defined in both documents.

6 Listing theorems

`\listofkeytheorems[⟨options⟩]`

Similar to `\listoffigures` or `\listoftables` but for theorems. For memoir and the AMS classes, `keytheorems` tries to copy the formatting of these commands as

defined by the class. For other classes, manual adjustments to `numwidth`^{→P.19} and `indent`^{→P.19} may be necessary.

`\keytheoremset{⟨options⟩}`

`\listofkeytheorems`

List of Theorems

1	Theorem	4
2	Theorem (Bertrand's postulate)	4
3	Theorem (Legendre's formula)	4
4	Theorem (Bézout's identity)	5
4*	Theorem	5
☛	Theorem	5
4	Theorem (continuing from p. 5)	5
4	Theorem (Bézout's identity, continuing from p. 5)	5
5	Theorem	6
6	Lemma	6
7	Theorem (Original)	7
1	Observation	7
2	Observation	7
1	Some Name	8
	Theorem	8
3.1	Conjecture	9
8	Lemma	9
1	Remark	9
1	Test	9
1	Example	10
1	Solution	10
1	Proposition	10
2	Proposition	10
A	Theorem	11
B	Theorem	11
1	Quote Theorem	12
1	Indented Theorem	12
1	Corollary	13
1	Definition (Dedekind domains)	13
2	Corollary (Cauchy's theorem)	13
1	Theorem (Quillen-Suslin)	14
2	Example	16
3	Example	16

6.1 Keys also defined in thmtools

`ignore={⟨comma-list of env names⟩}` (initially unset)

`ignoreall` (initially unset)

```

\listofkeytheorems[ignoreall,show=theorem]
\listofkeytheorems[
  ignoreall, show=conjecture,
  title=List of Conjectures
]

```

List of Theorems

1	Theorem	4
2	Theorem (Bertrand's postulate)	4
3	Theorem (Legendre's formula)	4
4	Theorem (Bézout's identity)	5
4*	Theorem	5
☛	Theorem	5
4	Theorem (continuing from p. 5)	5
4	Theorem (Bézout's identity, continuing from p. 5)	5
5	Theorem	6
7	Theorem (Original)	7

List of Conjectures

3.1	Conjecture	9
-----	----------------------	---

numwidth= $\langle length \rangle$ (initially 2.3em)

For the AMS classes, this is initially 1.5pc.

onlynamed= $\{\langle comma-list of env names \rangle\}$ (initially unset)

show= $\{\langle comma-list of env names \rangle\}$ (initially all theorems)

showall (initially set)

swapnumber=true|false (initially false)

title= $\langle text \rangle$ (initially `\GetTranslation{keythms_listof_title}`)

Defaults to “List of Theorems” if English or an unknown language is used. Currently several European languages have (likely inaccurate!) translations. A translation can be added with a GitHub pull request or manually with

`\DeclareTranslation{\lang}{keythms_listof_title}{\langle text \rangle}.`

6.2 Keys added by keytheorems

format-code= $\langle code with \#1, \#2, and \#3 \rangle$ (initially `\numberline{\#2}\#1\#3`)

Allows full control over the format for list entries. The theorem name is **#1**, the number is **#2**, and the (formatted) note is **#3**. The note formatting is still controlled by **note-code** ^{→ P. 20}.

indent= $\langle length \rangle$ (initially 1.5em)

Sets the left indent of items in the list of theorems. For memoir and the AMS classes, the indent is initially 0pt. It is not recommended to change this unless your class has different defaults not already covered.

`no-chapter-skip=true|false` (initially `false`)

By default a small vertical space is inserted between each chapter's chunk of theorems. Setting this key to `true` removes this space.

`chapter-skip-length=<length>` (initially 10pt)

Controls the amount of space inserted between chunks.

`no-continues=true|false` (initially `false`)

Suppresses the printing of theorems given the `continues`^{P.5} key in the list of theorems.

`no-title=true|false` (initially `false`)

Suppresses the title of the list of theorems. Useful for custom ordering of the list.

```
\keytheoremset{ignoreall}
\listofkeytheorems[show=example]
\listofkeytheorems[show=solution, no-title]
```

List of Theorems

1	Example	10
2	Example	16
3	Example	16
1	Solution	10

`no-toc=true|false` (initially `false`)

With the standard classes, lists of figures/tables are not added to the table of contents by default. The same is true for `\listofkeytheorems`, and with those classes this key does nothing. However some classes, notably `memoir` and the AMS classes, do add lists to the table of contents. With these classes, this key suppresses the addition of the list of theorems to the table of contents.

`note-code=<code with #1>` (initially `{ (#1)}`)

Formats the optional note in the list of theorems.

`onlynumbered={<comma-list of env names>}` (initially unset)

Similar to `onlynamed`^{P.19}, but lists only those theorems which are numbered. This is useful if you'd like to exclude things like unnumbered definitions and remarks from the list of theorems.

`print-body` (initially unset)

Instead of listing the theorem headings, the theorems are restated with their body text. Not very useful without the `store-all`^{P.3} load-time option.

`seq=<name>` (initially unset)

Used to list only the theorems added to the custom sequence `<name>` with the `seq`^{P.8} theorem key. This is the only way to fully customize which theorems appear in the list of theorems. Unlike with `show`^{P.19}, you do not need to use `ignoreall`^{P.18} to prevent theorems not in `<name>` from being printed.

`title-code=<code with #1>` (initially `\section*{#1}`)

If `\chapter` is defined, then initially this is instead `\chapter*{#1}`. This key has no effect if used with an AMS class because these classes hard-code the section heading into `\@starttoc`.

6.3 Adding code to list of theorems

There are analogous commands to `\addcontentsline` and `\addtocontents` for adding entries or arbitrary code to the list of theorems.

! You *must* use these commands rather than the aforementioned because the `.thlist` file is also used to define restated theorems and cannot contain unexpected code.

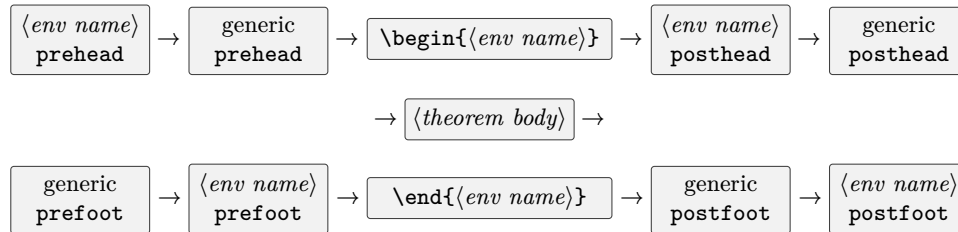
`\addtheoremcontentsline{<level>}{<text>}`

`\addtotheoremcontents{<code>}`

7 Theorem hooks

`\addtotheoremhook[<env name>]{<hook name>}{<code>}`

The `<hook name>` can be `prehead`, `posthead`, `prefoot`, `postfoot`, or `restated`. If no `<env name>` is given, the `<code>` is added to the “generic” hook, i.e. applied to all theorems. As in `thmtools`, the order of hooks is as follows:



The `restated` hook is applied at the start of theorems retrieved with the command `\getkeytheorem`, after the `prehead` hook. This can be useful for disabling commands such as `\footnote` in the restated theorems, e.g.

```
\addtotheoremhook{restated}{\renewcommand\footnote[2] [] {}}
```

By default, the `restated` hook disables the `\glossary`, `\index`, `\label`, and `\RecordProperties` commands.

In `thmtools`, the `prefoot` and `postfoot` hooks always prepend code, i.e. the code

```
\addtotheorempostfoothook{A}
\addtotheorempostfoothook{B}
```

results in BA after the theorem. With `keytheorems`, code is added in the order declared, meaning

```
\addtotheoremhook{postfoot}{A}
\addtotheoremhook{postfoot}{B}
```

results in AB after the theorem. This is the behavior of the `LATEX` kernel hooks that `keytheorems` uses under the hood.

Code added using the hook keys `preheadhook`^{P.9}, etc. is outermost, meaning executed first in `prehead` and `posthead` and last in `prefoot` and `postfoot`. Furthermore, if present, the `qed`^{P.10} symbol is placed *before* the `prefoot` hook.

8 Miscellaneous notes

8.1 beamer support

The package contains some *highly experimental* code to support theorems with beamer, including overlays. Most style keys are disabled by the default beamer theorem template. More become functional by setting

```
\setbeamertemplate{theorems}[ams style]
```

in the preamble. Alternatively, you have full control of theorems by setting the class option `noamsthm`.

Note that by default beamer defines a set of theorems when the class is loaded. These can be overwritten with `\renewkeytheorem`^{P.4} or disabled entirely with the `notheorems` class option.

Due to complications with overlays, writing contents of theorems to the `thlist` file is disabled. This means theorems can only be restated *after* their original statement. Furthermore, `\listofkeytheorems`^{P.17} is disabled and a warning issued if used.

User feedback is necessary to make this code fully compatible. Please report issues on the [Github page](#)!

8.2 Support for other classes

As mentioned in [section 4](#), the initial style key values set by `keytheorems` are adjusted for the AMS classes `amsart`, `amsbook`, and `amsproc`, the `amsart`-based `acmart` and `aomart`, and `jlreq`. You can find the exact changed values in the support files `keythms-⟨class⟩-support.tex`.

These class support files also contain code to adapt to class' formatting of lists-of as mentioned in [section 6](#); changes are made for the AMS classes, `memoir`, `IEEEtran`, and `jlreq`.

8.3 Support for font packages

Some font packages, all by Michael Sharpe, offer a `theoremfont` option that redefines the `plain` style body font to have italic text with upright figures, punctuation, and delimiters. `keytheorems` detects this option and sets its initial style values accordingly. The supported packages are `baskerville`, `cochineal`, `libertinust1math`, `newpxtext`, `newtx-text`, `scholax`, and `XCharter`.

8.4 Support for tagged PDF

The L^AT_EX team has been working hard to support the creation of tagged PDFs (see <https://latex3.github.io/tagging-project/>). The current `dev` formats make `amsthm` compatible with the kernel tagging code. Most of `keytheorems` is supported too, and anything that doesn't work should be reported. Explicitly not supported are the `tcolorbox`^{P.13} and `tcolorbox-no-titlebar`^{P.13} keys.

To produce a tagged PDF, add `\DocumentMetadata` in the first line of your document (additional instructions are found on the Tagging Project [website](#)). An example invocation might look like

```

\DocumentMetadata
{
  lang=en-US,
  pdfversion=2.0,
  pdfstandard=ua-2,
  tagging=on,
  % tagging-setup={math/setup=mathml-SE}, % optional
}

```

8.5 Public coding interfaces

`\l_keythms_thmuse_envname_tl`

Inside theorem environments and in all theorem hooks, you have access to the theorem’s environment and counter name in this token list variable.

`\keythms_getthm_theorem:nnnnn` $\{\langle name \rangle\}$ $\{\langle number \rangle\}$ $\{\langle restate counters \rangle\}$
 $\{\langle keys \rangle\}$ $\{\langle body text \rangle\}$

`\keythms_getthm_body:nnn` $\{\langle name \rangle\}$ $\{\langle restate counters \rangle\}$ $\{\langle body text \rangle\}$

Documentation coming soon.

`\keythms_listof_listcmd:nnnnnnn` $\{\langle name \rangle\}$ $\{\langle number \rangle\}$ $\{\langle Href \rangle\}$ $\{\langle page \rangle\}$
 $\{\langle restate counters \rangle\}$ $\{\langle keys \rangle\}$ $\{\langle body text \rangle\}$

Documentation coming soon.

`keytheorems/allthms/⟨hook name⟩`

`keytheorems/⟨envname⟩/⟨hook name⟩`

These are the “real” names for the hooks described in [section 7](#). They can be useful with `\AddToHookNext` or the kernel’s label mechanism for hooks.

9 Further examples

More examples will be added soon – rather, eventually... For now, you can find a `keytheorems` adaptation of `amsthm`’s classic `thmtest.tex` in the Github `tests` folder: `keytheorems-amsthmtest.tex`. There is also a version for tagged PDF: `tagged-keytheorems-amsthmtest.tex`.

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